Case Study

Institution: Shri Sant Gajanan Maharaj College of Engineering, Shegaon
Department: Electrical (Electronics & Power) Engineering
Course Title & Code: Electrical Machines – I (3EP02)
Class: Second Year Electrical Engineering
Semester: III
Course Teacher: Mr. B.S. Rakhonde
Title of the Innovative Practice: Use of YouTube Video Lectures for Enhanced Learning in Electrical Machines

Objectives / Goals of the Practice

1. **Enhance Conceptual Understanding of Magnetic Circuits**
   - Help students visualize the flow of magnetic flux, MMF, reluctance, and analogy with electric circuits.
2. **Enable Flexible & Self-Paced Learning**
   - Provide accessible YouTube video content for revision anytime, anywhere.
3. **Promote Visualization & Application**
   - Use animated explanations and problem-solving to strengthen clarity of core concepts.

Methods and Strategies Employed

- **Creation of Topic-Specific YouTube Video:**
  Developed an educational video focused on *Magnetic Circuit fundamentals*, types, laws, and numerical problem-solving.
- **Content Delivery Approach:**
  - Voice-over explanation with animated slides and diagrams.
  - Practical examples comparing magnetic circuits to electric circuits.
  - Step-by-step solution of sample problems.
- **Blended Teaching Method:**
  Students were asked to watch the video before class, followed by **interactive Q&A sessions** for deeper discussion.

Effective Presentation Strategy

1. **Classroom Integration:**
   - Links to the video were shared during lectures.
   - Follow-up discussions clarified doubts and strengthened understanding.
2. **Anytime Access:**
   - Students could revisit the video multiple times, especially before exams.

Program Outcomes (POs) & Program Specific Outcomes (PSOs) Mapped

- **POs:** PO1, PO2, PO3, PO4, PO12
- **PSOs:** PSO1

Reflective Critique

- **Initial Challenges:**
  Some students were new to learning through video-based resources and required guidance on how to utilize them effectively.
- **Improved Understanding:**
  Students reported better conceptual clarity of *Magnetic Circuits*, especially when solving numerical problems.
- **Faculty Feedback:**
  Suggested extending this approach to other fundamental topics like *Transformers* and *Induction Machines*.

Evidence of Success

- More than 50% of students viewed the video and participated actively in follow-up sessions.
- Students reported improved performance in class tests related to *Magnetic Circuits*.
- Many learners revisited the video multiple times for revision during exams.

Challenges Faced

- **Technical Issues:**
  Initial difficulties in video recording and editing.
- **Inclusivity Needs:**
  Requirement for subtitles/multilingual support to cater to diverse learners.

Supporting Links

- **Sample Video Link:** https://www.youtube.com/watch?v=0Ek08oPcjfs